$y = ax$

of masses of facts and figures
to draw valid conclusion.

Differential Equation — Ordinary (Ind. variable একটা 1 হয়)
— Partial ( " " " অনেক হয়)

# Independent variable
dependent "

# Find the differential eq$^n$ of the straight line $y = ax$

$$\frac{dy}{dx} = a$$

$$y = \frac{dy}{dx} \cdot x$$ → differential equation

$$y = Ae^{x} + Be^{-x}$$

$$\frac{dy}{dx} = Ae^{x} - Be^{-x}$$

$$\frac{d^2y}{dx^2} = Ae^{x} + Be^{-x}$$

$$\boxed{y = \frac{d^2y}{dx^2}}$$

Order → highest power
degree → " order এর power

$$\frac{d^2y}{dx^2} + xy\left(\frac{dy}{dx}\right)^2 = 0$$

$= 1$ → order 2, degree 1

Highest order এর এর power 1.

~~linear~~ → degree 1 হবে, ~~transgender~~ function থাকতে পারবে না, no prdouts of y or its derivate
~~non-linear~~

linear হতে হলে :
- degree 1 হবে
- transcendental function of y হতে না
- no products of y or its derivates.

$y^2$
$\frac{dy}{dx} y$
$\frac{d^2y}{dx} \cdot \frac{dy}{dx}$

i) $\frac{d^2y}{dx^2} + 5.\frac{dy}{dx} + 6y^2 = 0$ → no products of $y$ — order 2, degree 1, non linear, ordinary

ii) $\frac{d^2y}{dx^2} + 5y\left(\frac{dy}{dx}\right) + 6y = 0$ → no derivats of $y$ — order 2, degree 1, non linear, ordinary

iii) $\frac{d^2y}{dx^2} + y\sin x = 0$ — Order 2, degree 1, ~~non~~ linear, ordinary

iv) $\frac{d^4y}{dx^4} + x^2\frac{d^2y}{dx^2} + x^3\frac{dy}{dx} = xe^x$ — " 4, " 1, linear, "

v) $\frac{d^2y}{dx^2} + x\sin y = 0$ → transcendental of $y$ হবে না — " 2, " 1, nonlinear, "

# linear or non linear dependent variable এর উপর depend করে।